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# 324311 (14)

BE (3rd Semester)

Examination, April-May, 2013

Branch : Electrical, EEE

## MATHEMATICS - III

*Time Allowed : Three Hours*

*Maximum Marks : 80*

*Minimum Pass Marks : 28*

**Note :** Attempt all questions. Part (a) is compulsory in each question and attempt any two parts from remaining three parts in each question.

**Q. 1.** (a) Write Dirichlet's conditions for Fourier expansion of a function. 2

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(b) Obtain Fourier series for the function f(x) given by : 7

$$f(x) = \begin{cases} 1+\dfrac{2x}{\pi} & -\pi \le x \le 0 \\ 1-\dfrac{2x}{\pi} & 0 \le x \le \pi \end{cases}$$

Deduce that : $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\ldots=\frac{\pi^2}{8}$

(c) Expand f(x) = x sin x as a Fourier series in the interval $0 < x < 2\pi$. 7

(d) Obtain the constant term and coefficient of first sine and cosine terms in the Fourier expansion of y as given in the following table : 7

| | | | | | | |
|---|---|---|---|---|---|---|
| x : | 0 | 1 | 2 | 3 | 4 | 5 |
| y : | 9 | 18 | 24 | 28 | 26 | 20 |



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**Q. 2.** (a) Write the state of convolution theorem for inverse Laplace transform. 2

(b) Find the Laplace transform of the following : 7

(i) $f(t) = |t-1| + |t+1|, t \geq 0$

(ii) $\dfrac{\cos at - \cos bt}{t}$

(c) Find the inverse Laplace transform of the following : 7

(i) $\dfrac{s}{s^4 + 4a^4}$

(ii) $\dfrac{s^2}{(s^2 + a^2)(s^2 + b^2)}$

(d) Solve the differential equation : 7

$ty'' + 2y' + ty = \cos t$ given that $y(0) = 1$.



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**Q. 3.** (a) Obtain the partial differential equation by eliminating the arbitrary constant. 2

$$z = t(x^2 - y^2)$$

(b) Solve : 7

$$y^2p - xyq = x(z - 2y)$$

(c) Solve : 7

$$r + s - 6t = y \cos x$$

(d) Using method of separation of variables solve : 7

$$4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$$

given that $u = 3e^{-y} - e^{-5y}$ when $x = 0$.



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Q. 4. (a) Write the statement of residue theorem. 2

(b) Determine the analytic function $f(z) = u + iv$,

if : 7

$$u - v = \frac{\cos x + \sin x - e^{-y}}{2(\cos x - \cos hy)} \text{ and } f\left(\frac{\pi}{2}\right) = 0$$

(c) Evaluate $\int_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$ 7

where C is the circle $|z| = 3$.

(d) Show that : 7

$$\int_0^{2\pi} \frac{\cos 2\theta \, d\theta}{1 - 2a\cos\theta + a^2} = \frac{2\pi a^2}{1 - a^2} (a^2 < 1)$$



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Q. 5. (a) Write down the damping rule of z-transforms. 2

(b) Given that $z[u_n] = \frac{2z^2 + 3z + 4}{(z-3)^3}$

$|z| > 3$ show that $u_1 = 2$, $u_2 = 21$, $u_3 = 139$.

(c) Use convolution theorem to evaluate :

$$z^{-1}\left\{\frac{z^2}{(z-a)(z-b)}\right\}$$

(d) Solve the difference equation

$$y_{n+2} + 6y_{n+1} + 9y_n = 2^n$$

with $y_0 = y_1 = 0$ using z-transforms.



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